DISEASE PROJECT · GROUP 12DP-B

MODY2: Molecular Basis of a Monogenic Diabetes

Sequence, structure and cross-species comparison of human glucokinase (GCK)

GENERAL OBJECTIVE

To characterise how a disease-causing mutation in the human GCK gene reshapes the structure and function of glucokinase, and how this molecular change ultimately gives rise to the observable phenotype of GCK-MODY.

SPECIFIC OBJECTIVES

1. Retrieve and validate the reference and disease-associated GCK gene/protein sequences.
2. Characterise the effect of the assigned variant at sequence level.
3. Compare human GCK with Xenopus laevis and ≥ 1 further species; assess conservation.
4. Interpret the variant in the 3D structure of glucokinase.
5. Integrate the findings into a BMC-format report and defend it orally.

MILESTONES

M1	M2	M3	M4
Gantt chart delivered	Status retrospective	Final report submitted	Oral defence
09/10/26	**22/10/26**	**26/10/26**	**05/11/26**

WORK PACKAGES

WP	Name	Owner	Start	End
WP1	Repository & Setup	**Biel**	17/09/26	09/10/26
WP2	Literature Review	**Ari**	20/09/26	16/10/26
WP3	Bioinformatic Analysis	**Unai**	09/10/26	23/10/26
WP4	Report Writing	**Maeve**	14/10/26	26/10/26
WP5	Status Review & Defence	**Deolinda**	10/10/26	03/11/26
PROJECT	**Whole project**	—	**17/09/26**	**03/11/26**

PROJECT TEAM

Name	Role	Responsibility	Background
Biel	Repository & Setup Owner	Creates/maintains the GitHub repo; owns WP1 setup.	1st-year Bioinformatics student
Deolinda	Planning Lead	Owns the Gantt chart (v1 and the live v2 retrospective); owns WP5 (Status Review & Defence).	1st-year Bioinformatics student
Ari	Literature & Species Lead	Leads literature search and cross-species comparison; owns WP2 (Literature Review).	1st-year Bioinformatics student
Maeve	Analysis Support	Supports sequence/structure analysis and the written report; owns WP4 (Report Writing).	1st-year Bioinformatics student
Unai	Bioinformatic Analysis Lead	Leads sequence retrieval, variant and structural analysis; owns WP3 (Bioinformatic Analysis).	1st-year Bioinformatics student

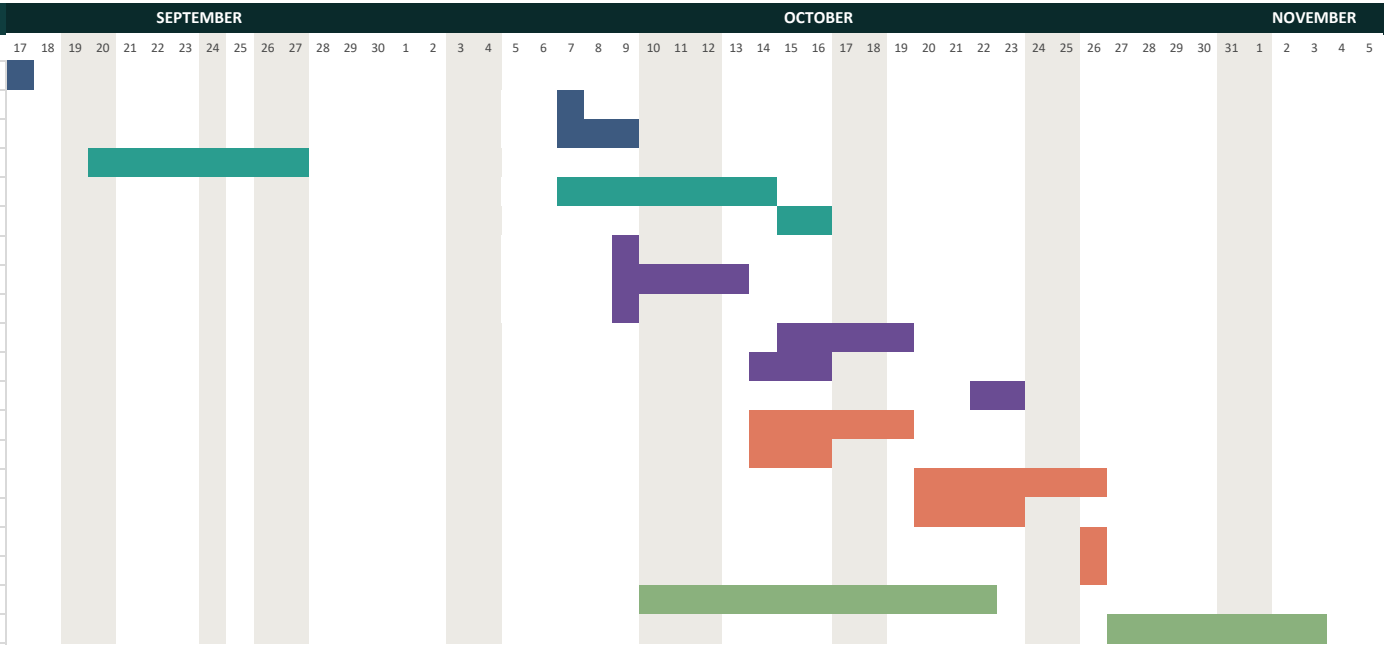
ID	Work Package	Task	Owner	Output	Duration (working	Pred.	Accomplished when	Comments/status	Start	End
1.1	WP1	Repository set-up and structure	Deolinda	Repo with folder structure	1	—	All members can push		17/09/26	17/09/26
1.2	WP1	Role and section assignment	All	Owner lines agreed	1	1.1	Every section has an owner	*Biel–repo/WP1, Deolinda–Gantt chart, Ari–literature, Maeve–analysis support, Unai–bioinformatics (all confirmed 07/10).*	07/10/26	07/10/26
1.3	WP1	Finish the gantt and deliver it	Biel	Gantt chart (PDF/XLSX) in /gantt folder	3	1.2	Delivered before 09.10 22:00		07/10/26	09/10/26
WP1 TOTAL									**17/09/26**	**09/10/26**
2.1	WP2	Initial paper scan	All	Shortlist of candidate papers	4	—	Each member has skimmed a few candidate papers on MODY2/GCK		20/09/26	27/09/26
2.2	WP2	Literature search on MODY2 and GCK	All	PDFs in references/	5	2.1	≥8 sources, reviews/originals separated	*Biel–pregnancy/GCK paper, Deolinda–MODY rapid review, Ari–citation review, Maeve–GCK mutations + precision diabetes papers, Unai–treatment options paper.*	07/10/26	14/10/26
2.3	WP2	Reference log with citation style	Deolinda	Reference log table	2	2.2	Every source cited in Mini Oxford style		15/10/26	16/10/26
WP2 TOTAL									**20/09/26**	**16/10/26**
3.1	WP3	Retrieve human GCK gene and protein	Maeve and Unai	Accession list + FASTA	1	—	Sequences match MODY2.zip		09/10/26	09/10/26
3.2	WP3	Select and justify the disease variant	Maeve and Unai	Variant sheet	2	3.1	Variant chosen with written rationale		09/10/26	13/10/26
3.3	WP3	Select comparison species	Ari and Biel	Species table with justification	1	3.1	X. laevis included, others justified		09/10/26	09/10/26
3.4	WP3	Pairwise and multiple sequence alignment	Maeve	Alignment files + figure	3	3.3	Conservation of mutated position assessed	*Maeve: retrieves orthologous sequences for X. laevis + 2nd species (15/10), pairwise BLAST alignment vs. human GCK (16/10), then MSA + conservation figure (19/10).*	15/10/26	19/10/26
3.5	WP3	Structural analysis (PDB / AlphaFold)	Unai	Structure figure	3	3.2	Variant position located in structure		14/10/26	16/10/26
3.6	WP3	Structure superposition healthy vs variant	Ari	Superposition figure	2	3.5	Structural consequence described		22/10/26	23/10/26
WP3 TOTAL									**09/10/26**	**23/10/26**

ID	Work Package	Task	Owner	Output	Duration (working	Pred.	Accomplished when	Comments/status	Start	End
4.1	WP4	Background section	Ari	Background section	4	2.2	Within agreed length, fully cited	*Ari drafts disease background first (14–15/10), then gene/protein background (16–19/10).*	14/10/26	19/10/26
4.2	WP4	Methods section	Biel	Methods section	3	3.2	Reproducible: sources, tools, parameters		14/10/26	16/10/26
4.3	WP4	Results section	Maeve	Results section (≤5 figures)	5	3.4	Findings only, no interpretation	*Maeve drafts sequence/variant results first (20–21/10), then alignment/conservation results (22–23/10), then structural results once the 22.10 lab is done (26/10).*	20/10/26	26/10/26
4.4	WP4	Discussion and Conclusions	Unai	Discussion & Conclusions	4	3.4	Answers the project question	*Unai interprets sequence + alignment findings together (20–21/10), then relates them to clinical phenotype and states limitations (22–23/10).*	20/10/26	23/10/26
4.5	WP4	Abstract and keywords	Deolinda	Abstract (≤350 words)	1	4.3, 4.4	Background/Results/Conclusions structure		26/10/26	26/10/26
4.6	WP4	Format and citation review	All	Final report.md	1	4.5	≤7 pages/5000 words, ≤5 figures, style ok	*Each member reviews their own section: Deolinda–Overview/Abstract, Maeve–Methods/Gantt, Ari–Background/citations, Unai–Discussion/Conclusions, Biel–Results/figures.*	26/10/26	26/10/26
WP4 TOTAL									**14/10/26**	**26/10/26**
5.1	WP5	Live Gantt update and retrospective	Biel	Gantt v2 + gap analysis	8	—	Differences v1 vs v2 explained; tracked continuously since M1		10/10/26	22/10/26
5.2	WP5	Defence preparation, all sections	All	Shared defence notes	6	4.6	Every member can explain every section	*Deolinda–Overview/Background, Maeve–Methods/Gantt retrospective, Ari–species comparison/conservation, Unai–Discussion/Conclusions, Biel–Results/structural analysis.*	27/10/26	03/11/26
WP5 TOTAL									**10/10/26**	**03/11/26**
PROJECT TOTAL									**17/09/26**	**03/11/26**

Durations are given in working days (Mon–Fri); weekends and local holidays (24/09, 12/10 and 01/11) are excluded.

PROJECT TIMELINE

ID	Task	Owner
1.1	Repo with folder structure	Deolinda
1.2	Owner lines agreed	All
1.3	Gantt chart (PDF/XLSX) in /gantt folder	Biel
2.1	Shortlist of candidate papers	All
2.2	PDFs in references/	All
2.3	Reference log table	Deolinda
3.1	Accession list + FASTA	Maeve and Unai
3.2	Variant sheet	Maeve and Unai
3.3	Species table with justification	Ari and Biel
3.4	Alignment files + figure	Maeve
3.5	Structure figure	Unai
3.6	Superposition figure	Ari
4.1	Background section	Ari
4.2	Methods section	Biel
4.3	Results section (≤5 figures)	Maeve
4.4	Discussion & Conclusions	Unai
4.5	Abstract (≤350 words)	Deolinda
4.6	Final report.md	All
5.1	Gantt v2 + gap analysis	Biel
5.2	Shared defence notes	All



Milestones

Legend:

WP1 | W P2 | W P3 | W P4 | W P5 | *weekend/holiday*

RISK MATRIX

Impact → / Likelihood ↓

	I=1	I=2	I=3
L=3			R1
L=2		R2 R3 R5	
L=1			R4

ID	Risk	Why	L	I	Score	Category	Contingency
R1	Superposition lab (22.10) leaves very little buffer before the final report deadline (26.10)	Because the superposition technique is only taught in the 22/10 lab (separate from the 01/10 PDB lab), only working days remain before the 26/10 deadline	3	3	9	High	Draft Results/Discussion with alignment data first (ready 19.10); keep the structural superposition as a block added last, once the lab is done
R2	Merge conflicts in the shared GitHub repository when several members edit the same files at once	Because 5 people contribute to report.md and related files without a strict branching workflow, concurrent edits are likely to collide	2	2	4	Medium	One file/section per person; use pull requests for review; resolve conflicts promptly as a team instead of editing the same file simultaneously
R3	No experimental PDB structure available for the chosen variant	Because not every GCK variant has a solved experimental structure deposited in the PDB	2	2	4	Medium	Use an AlphaFold prediction and state the limitation explicitly in the report
R4	Sequence retrieved does not match the provided MODY2.zip	Because public databases (NCBI/Ensembl) may use a different reference transcript/isoform than the one provided in MODY2.zip	1	3	3	Medium	Use MODY2.zip sequences as reference and document the discrepancy
R5	A member is ill or unavailable during the final week	Because the final week concentrates several dependent deliverables (report, defence prep), so any absence has high impact	2	2	4	Medium	All work lives on GitHub from day one, so anyone can pick up a section

Score = Likelihood × Impact | Low < 3 Medium 3–5 High > 5